Russell Dudek

INTERVIEW THESIS BRIEF · PRODUCT DIRECTOR, PHYSICAL AI PLATFORM

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EXECUTIVE BRIEF · 01

Make physical work addressable.

DeepHow's next platform advantage is one governed work address that makes knowledge, expected action, observed execution, evidence, and learning interoperable across the product portfolio.

The mandate behind the title

Turn four high-value product lines into one legible platform that compresses first value, improves trust, supports enterprise requirements, and compounds learning from every deployment.

WHY NOW

Product expansion

DeepHow's public platform now spans knowledge capture, on-demand output verification, live SOP verification, and time-and-motion intelligence.

Physical AI proof

The Foxconn/NVIDIA work publicly demonstrates live video reasoning tied to procedural adherence and first-pass-yield improvement.

Platform pressure

More models, evidence types, product surfaces, integrations, and customer commitments increase the cost of unclear shared contracts.

Commercial leverage

A common work identity and evidence system can reduce activation friction and make expansion across product surfaces more coherent.

CENTRAL CONTRADICTIONS

Standardization vs. legitimate variation · Experiment speed vs. operational trust · Enterprise controls vs. frontline usability · Strategic-account commitments vs. reusable platform leverage · Model capability vs. explainable customer outcomes.

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OPERATING ARCHITECTURE · 02

One performed unit of work becomes the platform's common object.

COORDINATE 1	COORDINATE 2	COORDINATE 3	COORDINATE 4	COORDINATE 5
Work identity	Expected action	Observed execution	Evidence contract	Learning return

PLATFORM IMPLICATIONS

Ontology and event model

Define durable identifiers and relationships for process, station, asset, role, SOP version, step, output, event, cycle, exception, and result.

Evidence contract

Make confidence, latency, precision/recall posture, tolerance, human review, fallback, and recovery explicit by decision consequence.

Service boundaries and APIs

Expose stable platform capabilities while preventing every product line or customer integration from creating a separate interpretation of work.

Security and data lifecycle

Control identity, permissions, residency, retention, model access, traceability, and customer data rights across video, photo, documents, and derived events.

Model and evaluation lifecycle

Version data, models, prompts, thresholds, and evaluation sets; observe drift, overrides, and outcome performance after release.

Builder experience

Make the platform understandable through schemas, SDK/API patterns, sandbox data, observability, error states, and fast paths to a useful result.

Standardize where trust and leverage depend on consistency.

Federate where industry, workflow, environment, model, or risk creates legitimate variation. Record the decision so exceptions do not quietly become architecture.

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DECISION SYSTEM · 03

AI thresholds are product decisions, not model footnotes.

THRESHOLD QUESTIONS BY USE CASE

DECISION DIMENSION	KNOWLEDGE CAPTURE	OUTPUT VERIFICATION	LIVE SOP VERIFICATION	TIME & MOTION
Consequence	Wrong or stale guidance	Incorrect acceptance/ rejection	Missed or late intervention	Misclassified work or misleading optimization
Latency	Retrieval-time useful	Task-completion useful	Real-time / step-time critical	Analysis-cycle useful
Human authority	Expert approves knowledge	Reviewer handles exceptions	Operator/supervisor retains action authority	Engineer validates improvement decision
Evidence	Source, version, scope	Reference, tolerance, confidence	Sequence, timing, context, alert	Cycle, class, variation, sample coverage
Recovery	Correct and republish	Re-review and disposition	Escalate, pause, or safe fallback	Reclassify and rerun analysis

IN-MARKET EXPERIMENT DISCIPLINE

Frame

Name the customer annoyance, intended outcome, current baseline, affected users, and why a platform intervention is justified.

Bound

Define data rights, model scope, authority, thresholds, fallback, owner, duration, and stop conditions.

Instrument

Measure first value, use, model behavior, human override, workflow outcome, support burden, and reusable learning.

Decide

Formalize, revise, federate, or stop based on evidence—not executive enthusiasm or sunk cost.

MONTHLY SCORECARD

Activation speed · Reusable activation · Decision reliability · Workflow pull · Agreed operational outcome · Cost per useful outcome · Strategic expansion · Technical and support burden.

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INTERVIEW THESIS BRIEF · CANDIDATE VALUE AND DISCUSSION

CONTRIBUTION PROFILE · 04

A platform leader who can read the factory, the architecture, and the operating system.

VERIFIED CONTRIBUTION MAP

Physical operations

Manufacturing, quality, standard work, Lean/Gemba, throughput, high-reliability electronics, and frontline execution.

Software-enabled machines

Current robotics and machine-vision environment, fleet feedback, telemetry, support/engineering issue flow, regulated drones, and computer-vision concepts.

AI workflow systems

AI readiness, RAG, agentic patterns, evaluation, human authority, escalation, governance, adoption, and operational value.

Enterprise translation

Connects engineers, operators, quality, customers, executives, systems, partners, and business outcomes without flattening technical tradeoffs.

Affirmative platform-authority case

Russell's advantage is system-level product judgment grounded in direct operating work: defining the work, making signals useful, clarifying decisions, installing mechanisms, and ensuring technology produces an adopted outcome.

EXECUTIVE QUESTIONS

1. What customer outcome must become easier because this role exists?
2. Where are product-line differences legitimate—and where are they symptoms of missing shared contracts?
3. Which confidence, latency, and human-authority decisions are currently hardest for customers?
4. What builder or frontline annoyance most often delays first value?
5. Which deployment learning is not yet returning to the platform?
6. Where does roadmap authority become unclear among customer commitments, product, and architecture?

PROPOSED INTERVIEW WORKING SESSION

Take one current cross-product customer journey. Map its work identity, expected action, observations, evidence contract, learning return, platform dependencies, and decision owners. Use the exercise to test the Work Address hypothesis against DeepHow's actual architecture and customer reality.